

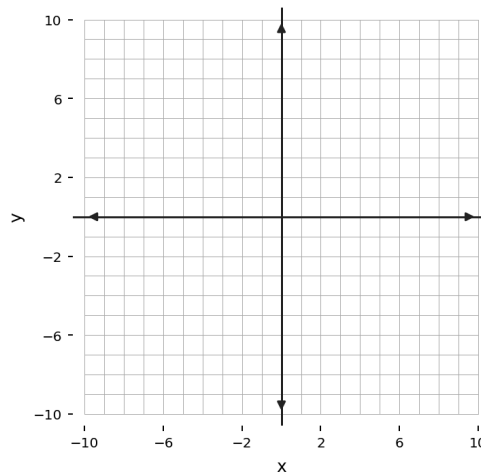
Algebra 1 · Unit 1 Teacher Practice

Coordinate-Plane Graphs · One-Off Practice Packet

Teacher Key. Answers and concise solutions follow each item.

1. I Can: Plot and read ordered pairs.

Plot the point R(5,-7) on the coordinate plane. Label the point R.



Answer: R is at (5, -7).

Solution: Move 5 units right and 7 units down, then label the point R.

2. I Can: Plot and read ordered pairs.

Start at the origin. Move 4 units left and then 7 units down. Write the ordered pair for the ending point.

Answer: (-4, -7)

Solution: Left makes the x-coordinate -4; down makes the y-coordinate -7.

3. I Can: Plot and read ordered pairs.

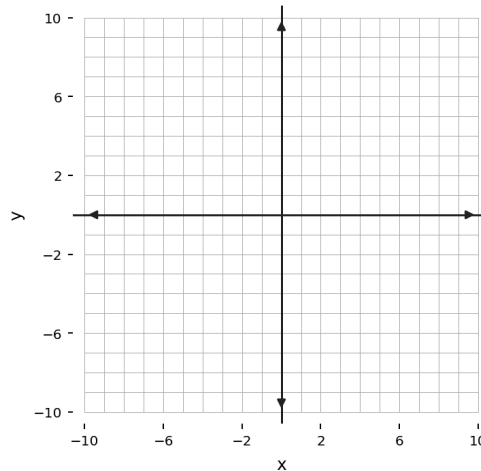
A student says, “I moved 5 units right and 4 units down from the origin, so the point is (-5,-4).” Explain the error and write the correct ordered pair.

Answer: (5, -4); the x-coordinate must be positive.

Solution: Moving right gives a positive x-coordinate, so the correct point is (5, -4).

4. I Can: Plot and read ordered pairs.

Plot the point $S(0, -6)$ on the coordinate plane. Label the point S.

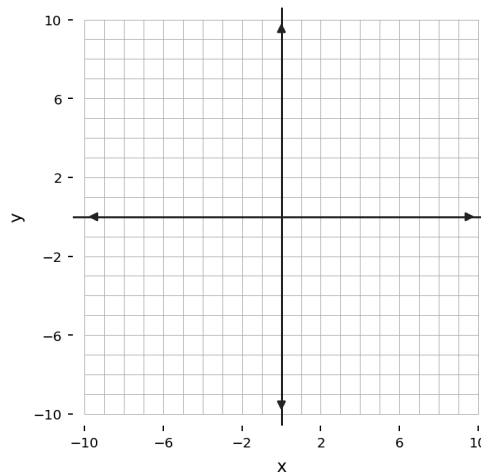


Answer: S is at (0, -6).

Solution: Because $x=0$, the point lies on the y-axis; move 6 units down and label S.

5. I Can: Use the x- and y-axes correctly.

For $T(-3, 5)$, state which axis controls the -3 movement and which axis controls the 5 movement. Then plot T.



Answer: x-axis: 3 left; y-axis: 5 up; $T(-3, 5)$.

Solution: The first coordinate controls horizontal movement on the x-axis; the second controls vertical movement on the y-axis.

6. I Can: Use the x- and y-axes correctly.

A graph will show how the amount of water in a tank changes as minutes pass. Which quantity belongs on the x-axis and which belongs on the y-axis? Give a one-sentence reason.

Answer: Minutes on the x-axis; amount of water on the y-axis.

Solution: Time is the input that progresses, and the amount of water depends on that time.

7. I Can: Use the x- and y-axes correctly.

A student plots $W(4,-2)$ by moving 4 units down and 2 units right. Explain the axis mistake and describe the correct moves.

Answer: Move 4 right, then 2 down.

Solution: The x-coordinate 4 controls horizontal movement; the y-coordinate -2 controls vertical movement.

8. I Can: Use the x- and y-axes correctly.

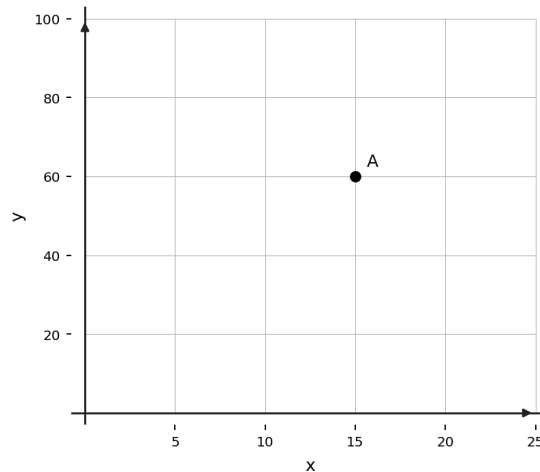
A graph will show total money collected as the number of tickets sold increases. Which quantity belongs on each axis? Explain briefly.

Answer: Tickets sold on x-axis; money collected on y-axis.

Solution: The number of tickets sold is the input; the money collected depends on the number sold.

9. I Can: Read a graph using its scale.

Use the scale on the graph to write the ordered pair for point A.

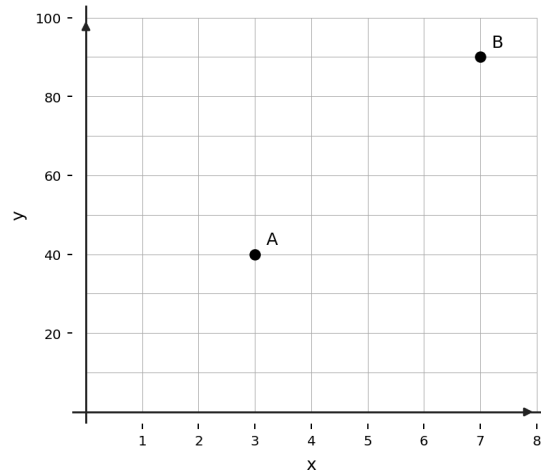


Answer: (15, 60)

Solution: The x-axis counts by 5 and the y-axis counts by 20. A is at $x=15$ and $y=60$.

10. I Can: Read a graph using its scale.

Use the graph scale to read the y-values of A and B. By how much is B's y-value greater than A's?

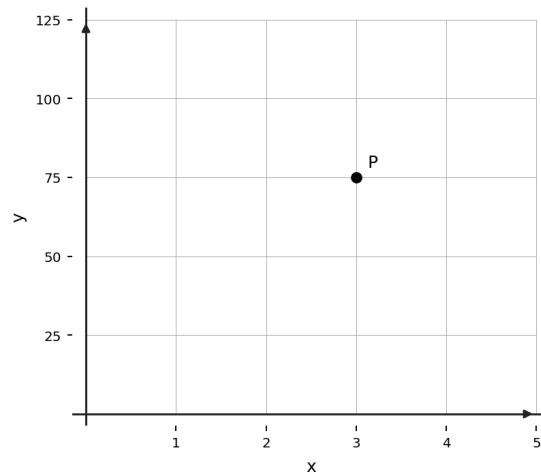


Answer: 50

Solution: A has $y=40$ and B has $y=90$, so $90-40=50$.

11. I Can: Read a graph using its scale.

Point P is shown on the graph. A student says P has y-coordinate 3 because it is three gridlines above 0. Explain the error and give P's correct y-coordinate.

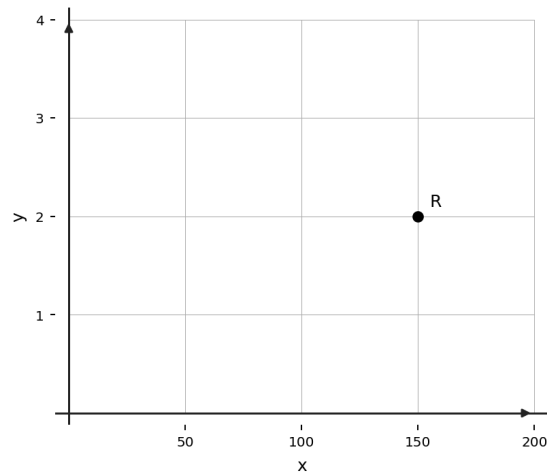


Answer: 75

Solution: The y-axis increases by 25 per gridline, so three gridlines above 0 is 75.

12. I Can: Read a graph using its scale.

Point R is shown on the graph. A student says its x-coordinate is 3 because it is on the third vertical gridline to the right of 0. Explain the error and give the correct x-coordinate.

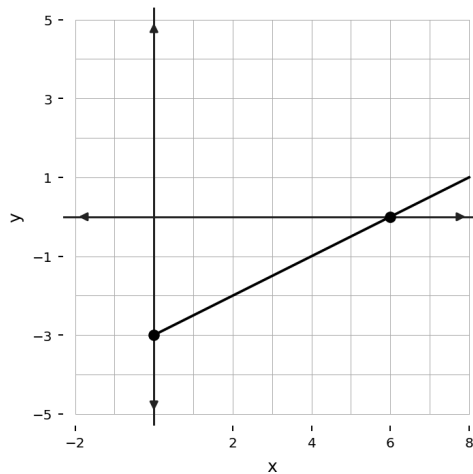


Answer: 150

Solution: The x-axis increases by 50 per gridline, so the third gridline is $x=150$.

13. I Can: Identify and interpret intercepts when they are shown.

Identify the x-intercept and the y-intercept of the line shown.

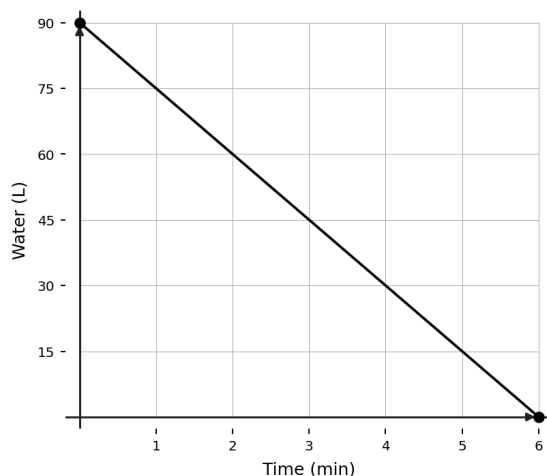


Answer: x-intercept (6,0); y-intercept (0,-3).

Solution: The x-intercept is where $y=0$; the y-intercept is where $x=0$.

14. I Can: Identify and interpret intercepts when they are shown.

The graph shows water remaining in a tank while it drains. Interpret the x-intercept $(6,0)$ in this situation.

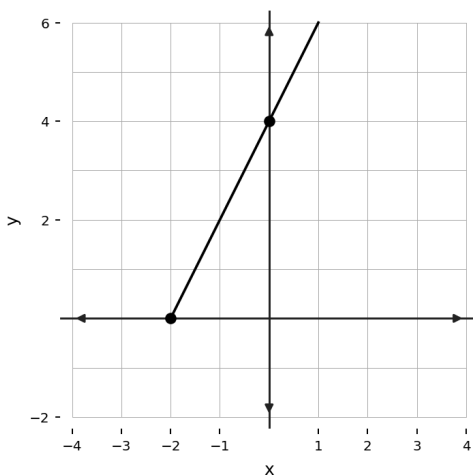


Answer: After 6 minutes, 0 liters remain; the tank is empty.

Solution: At an x-intercept, the y-value is 0. Here x is time and y is water remaining.

15. I Can: Identify and interpret intercepts when they are shown.

A student says, “The point $(0,4)$ is the x-intercept because its x-coordinate is 0.” Explain the error. Then identify both intercepts of the line.



Answer: $(0,4)$ is the y-intercept; x-intercept $(-2,0)$, y-intercept $(0,4)$.

Solution: An x-intercept has $y=0$. A y-intercept has $x=0$.
